

10(a)	any seven from: <i>selective reabsorption – sodium ions</i> 1 active transport of Na⁺, out of (pct) cells / into blood ; 2 Na⁺ ion concentration, gradient (produced) / reduced in cell ; 3 Na⁺ enters (pct) cells from, lumen / tubule / filtrate ; 4 by facilitated diffusion / using carrier protein ; 5 cotransport of, glucose / amino acids / ions ; 6 glucose diffuses into blood ; <i>synapse – calcium ions</i> 7 (when) presynaptic membrane depolarised ; 8 calcium (ion) channels / voltage-gated channels, open ; 9 calcium ions enter presynaptic neurone ; 10 stimulate vesicles of ACh to, move towards / fuse with, presynaptic membrane ; 11 causing exocytosis of ACh ;	7
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Question	Answer					Marks
10(b)	<i>any eight from:</i>					8
	<i>differences</i>					
			<i>nervous</i>		<i>endocrine</i>	
	1	communication	action potential / impulse	and	hormone ;	
	2	nature of communication	electrical (and chemical)	and	chemical ;	
	3	mode of transmission	neurone / nerve cell	and	blood ;	
	4	response destination	muscle / gland	and	target, organs / tissue / cells ;	
	5	transmission speed	fast(er)	and	slow(er) ;	
	6	effects	specific / localised	and	(can be) widespread ;	
	7	response speed	fast(er)	and	slow(er) ;	
	8	duration	short-lived / temporary	and	can be long-lasting / permanent ;	
	9	receptor location	on cell surface membrane	and	either on cell surface membrane or within cell ;	
	<i>similarities</i>					
	10	cell signalling	both involve cell signalling ;			
	11	detail	both involve signal molecule binding to receptor ;			
12	chemicals	both involve chemicals ;				